

An aerial photograph of a coastal region. The top half of the image shows a green, forested land area with numerous small white buildings scattered throughout. A dark, winding river or stream flows through the land towards the bottom. The bottom half of the image shows a body of water with varying shades of blue and green, indicating different depths or water quality. The coastline is visible as a thin line separating the land from the water.

Data Lunch 2025

Spencer Rhea

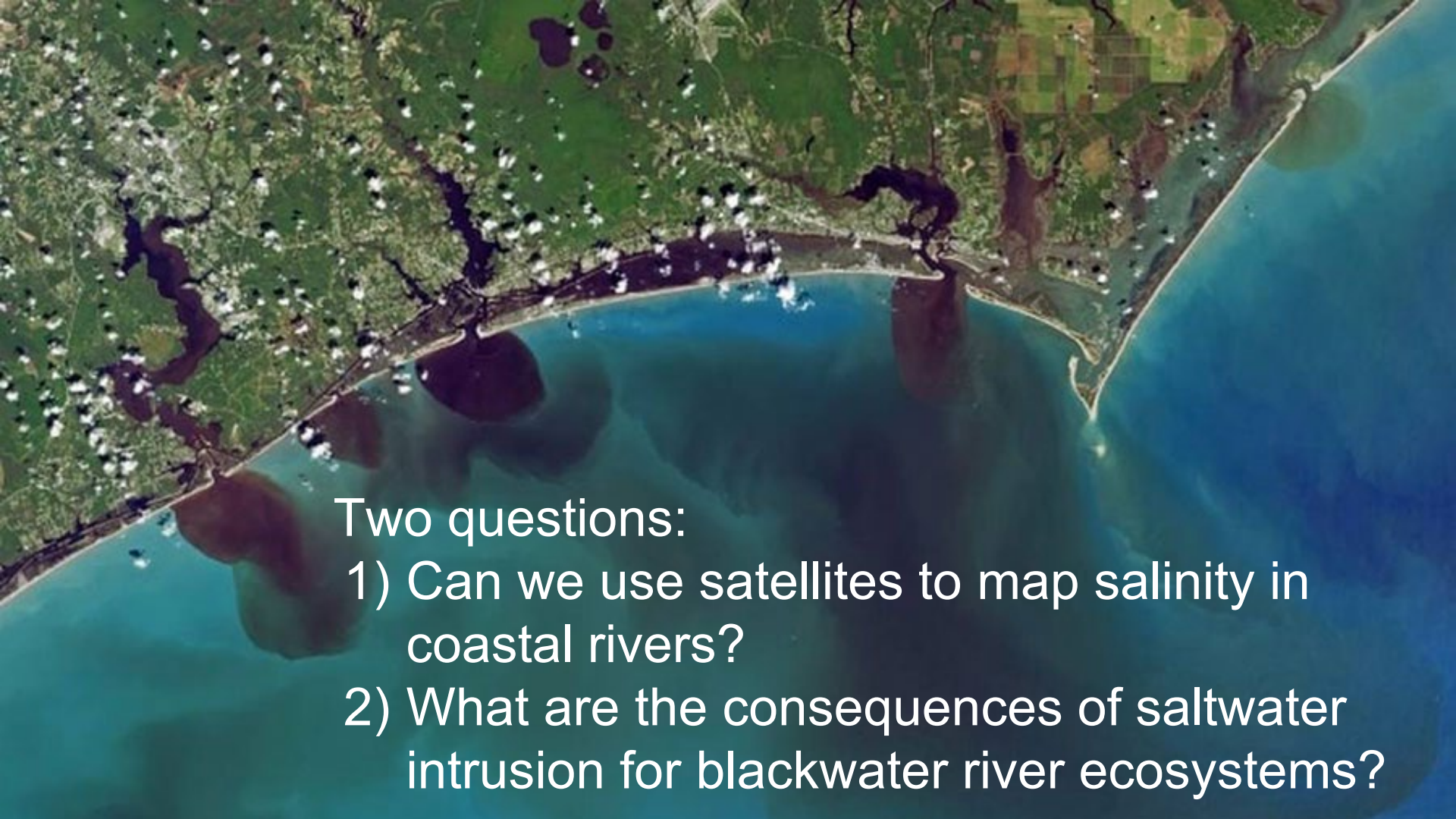


Blackwater Rivers: Rivers with a high concentration of dissolved organic carbon

An aerial photograph of a coastal region. The land is green with some urban areas and roads. A large body of water, likely a bay or estuary, is visible, with a prominent dark, winding waterway. The coastline is irregular, with a sandy beach area. The water transitions from a shallow turquoise near the shore to a deeper blue further out.

We know sea levels are rising, hurricanes are strengthening, and precipitation is becoming more variable

These all lead to **saltwater intrusion**



Two questions:

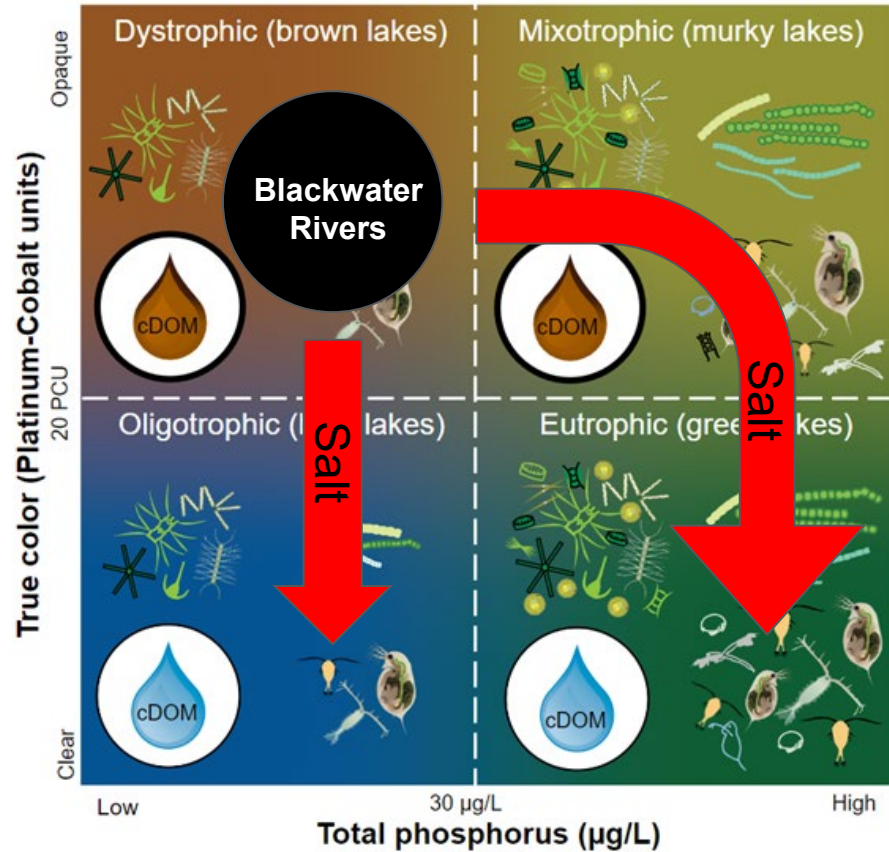
- 1) Can we use satellites to map salinity in coastal rivers?
- 2) What are the consequences of saltwater intrusion for blackwater river ecosystems?



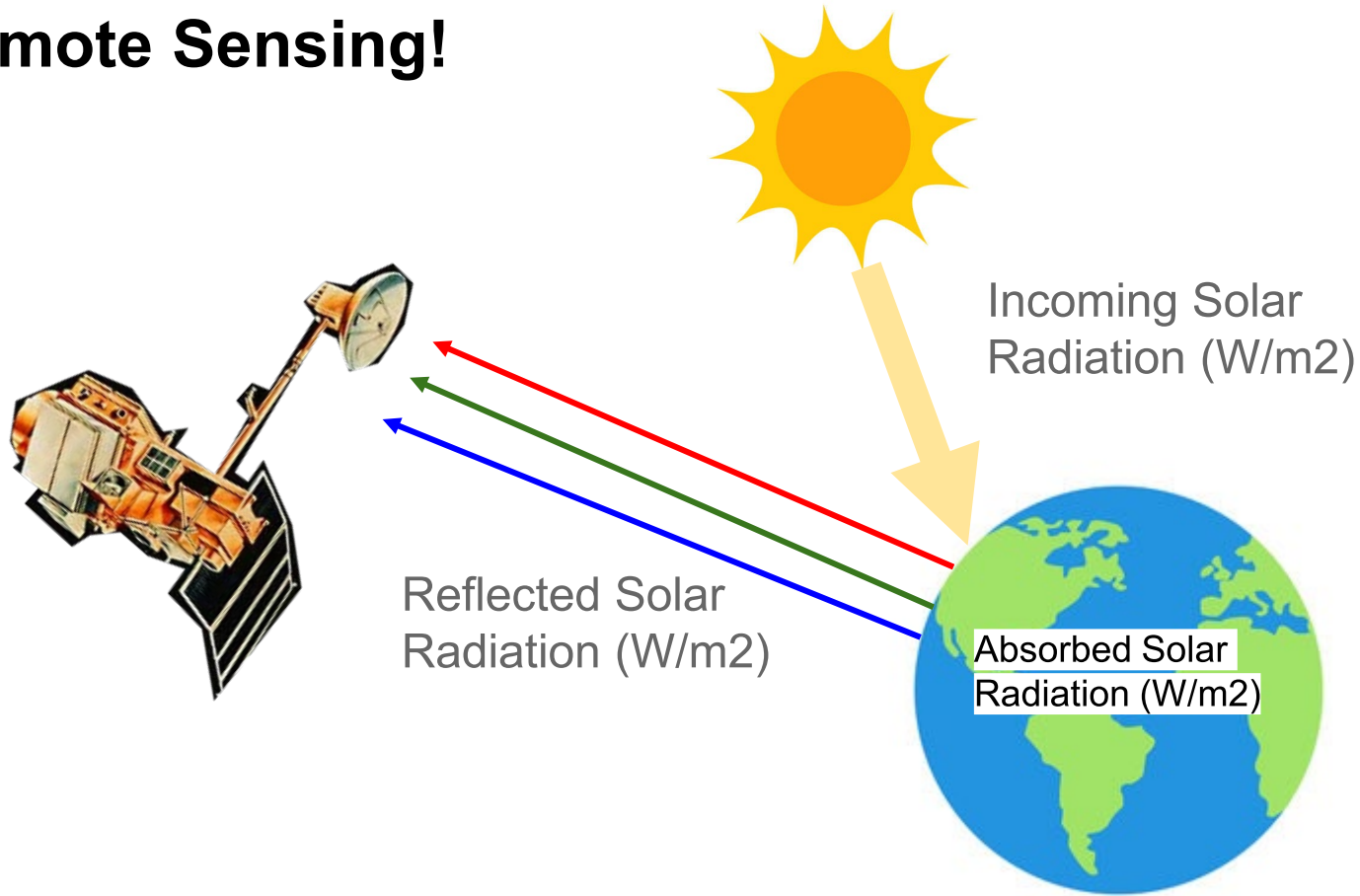
1) Can we use satellites to map salinity in coastal rivers?

2) What are the consequences of saltwater intrusion for blackwater river ecosystems?

How do we answer this questions with *(free and publicly available)* data?



Remote Sensing!



$$\text{Reflectance (\%)} = (\text{Reflected Solar Radiation} / \text{Incoming Solar Radiation}) * 100$$

Remote Sensing: What can it tell us about water?

Colored Dissolved
Organic Matter
(cDOM)



Turbidity/suspended
solids



Chlorophyll-a
(Chl-a)

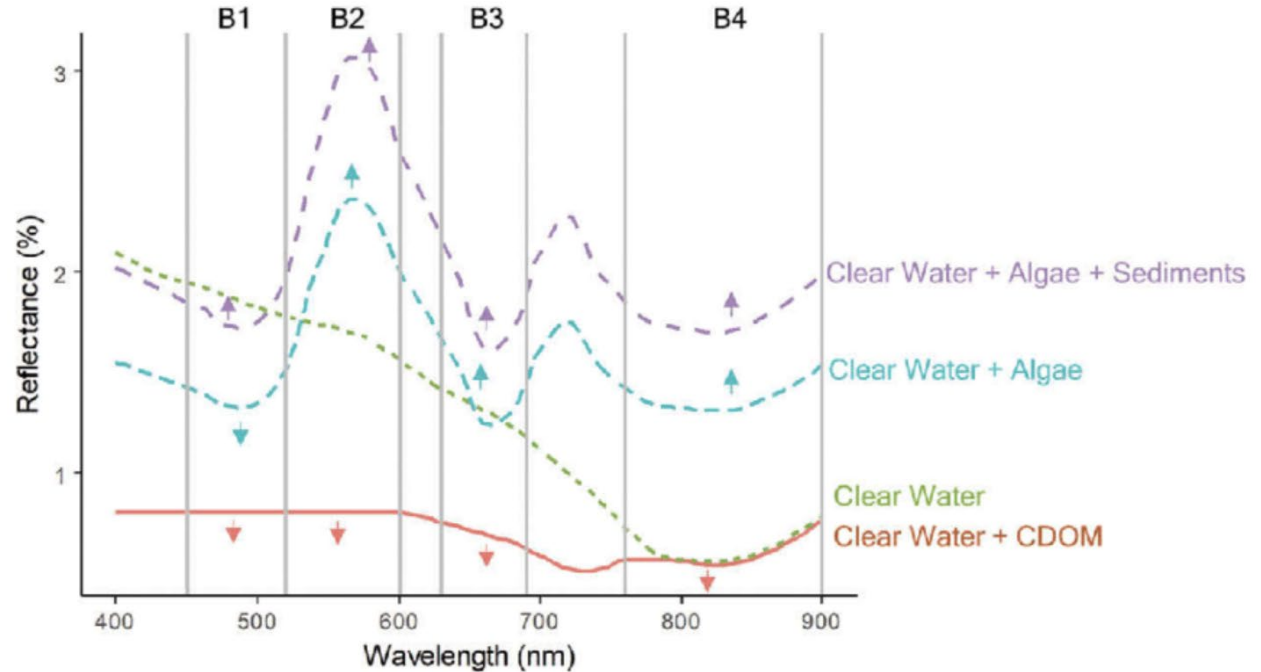


Water Extent



And Temperature!

Remote Sensing: Why can it tell us things about water?



Schematic diagram of water reflectance affected by algae, sediments, and CDOM (coloured dissolved organic matter). Rectangles inside indicate the Landsat TM/ETM+ band positions of Band 1 (B1), Band 2 (B2), Band 3 (B3), and Band 4 (B4). Arrows indicate the expected changes in the curve when concentrations of corresponding substances increase (based on data from Carder et al. (1989) and Luoheng Han (1997)).

I need three kinds of data

1. Satellite imagery dataset
2. River network dataset
3. Salinity dataset



Join them
all together!

National Hydrography Dataset

Landsat Missions: Imaging the Earth Since 1972

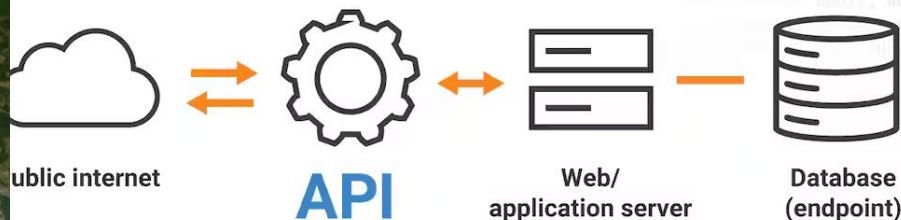
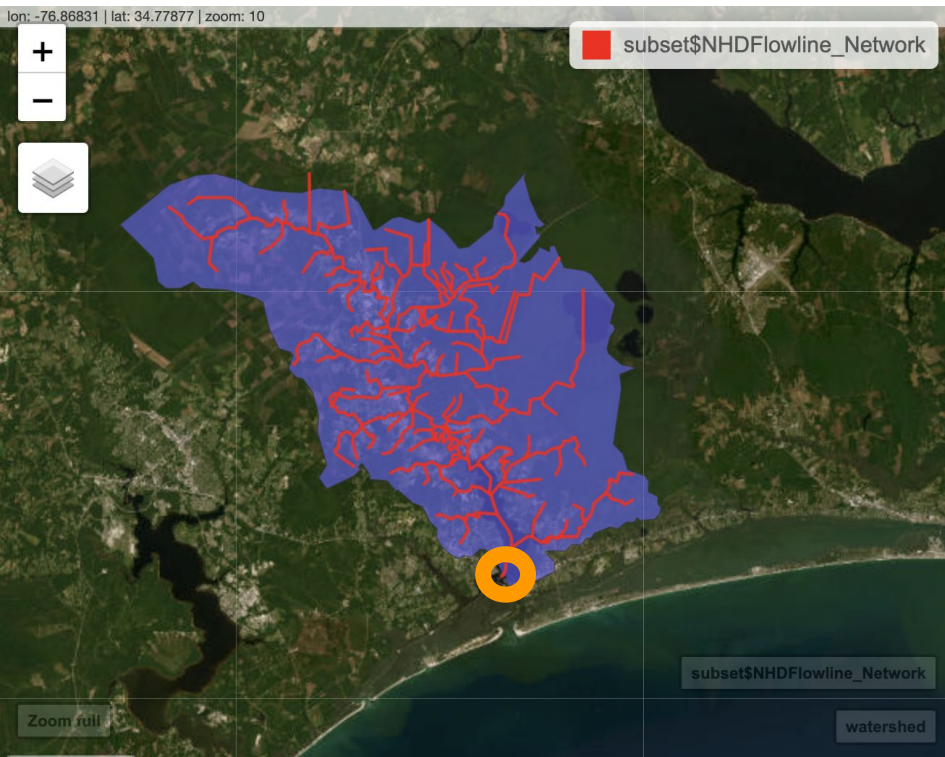


National Water Quality Portal



Acquiring data: National Hydrography Dataset

R Package + API



Acquiring data: Satellite Data (Landsat)



Landsat 5,7,8,9
and Harmonized
Landsat-Sentinel
Image



Fmask
Cloud Mask



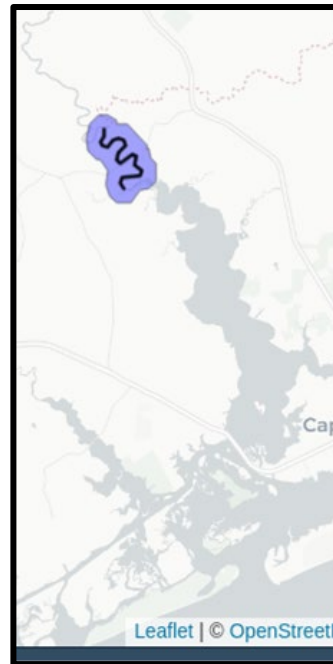
Google Earth Engine



Fmasks/DSWE
Water Mask



Only Water



Extract Color
for NHD
Flowlines

Acquiring data: Salinity data

R Package + API !

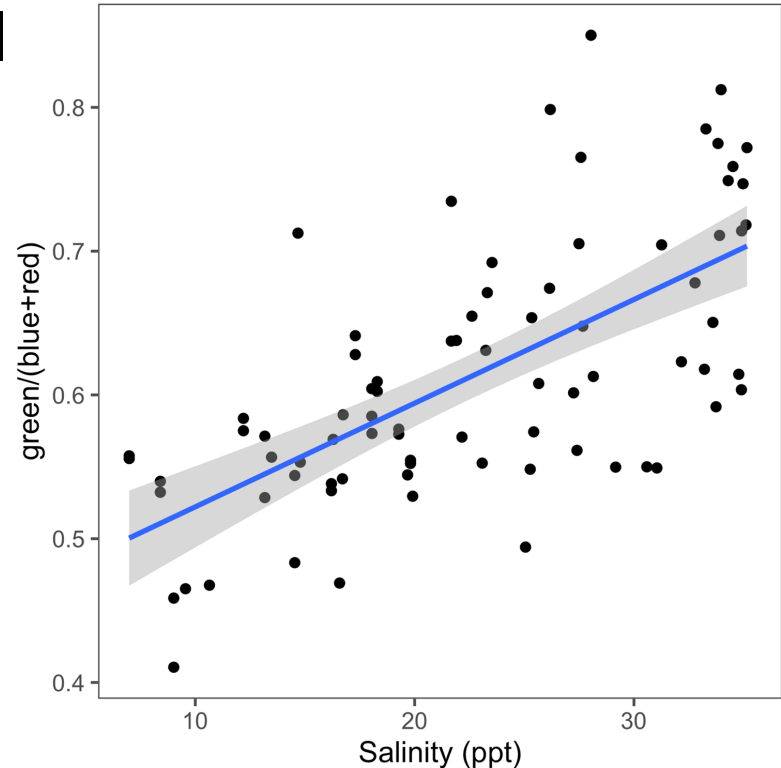


Acquired all the data (yay)!

Can we answer our original questions?

1) Can we use satellites to map salinity in coastal rivers?

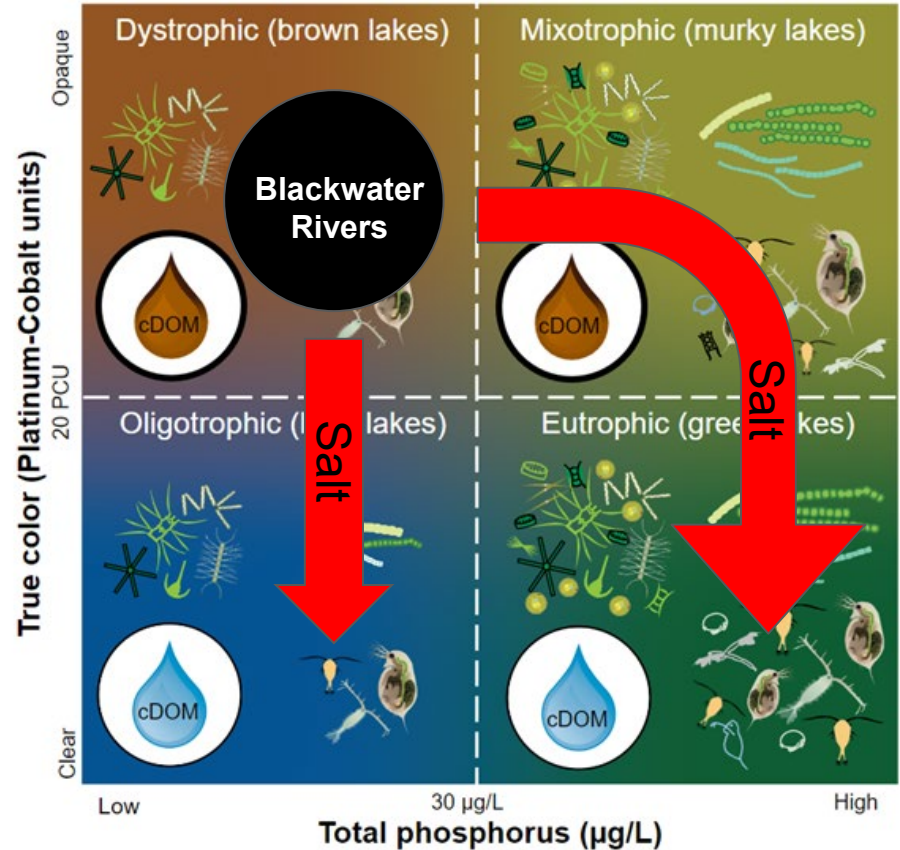
- a. Does dissolved organic matter and salinity have the expected relationship?
- b. Can we see this relationship with satellite imagery?



Acquired all the data (yay)!

Can we answer our original questions?

2) What are the consequences of saltwater intrusion for blackwater river ecosystems?



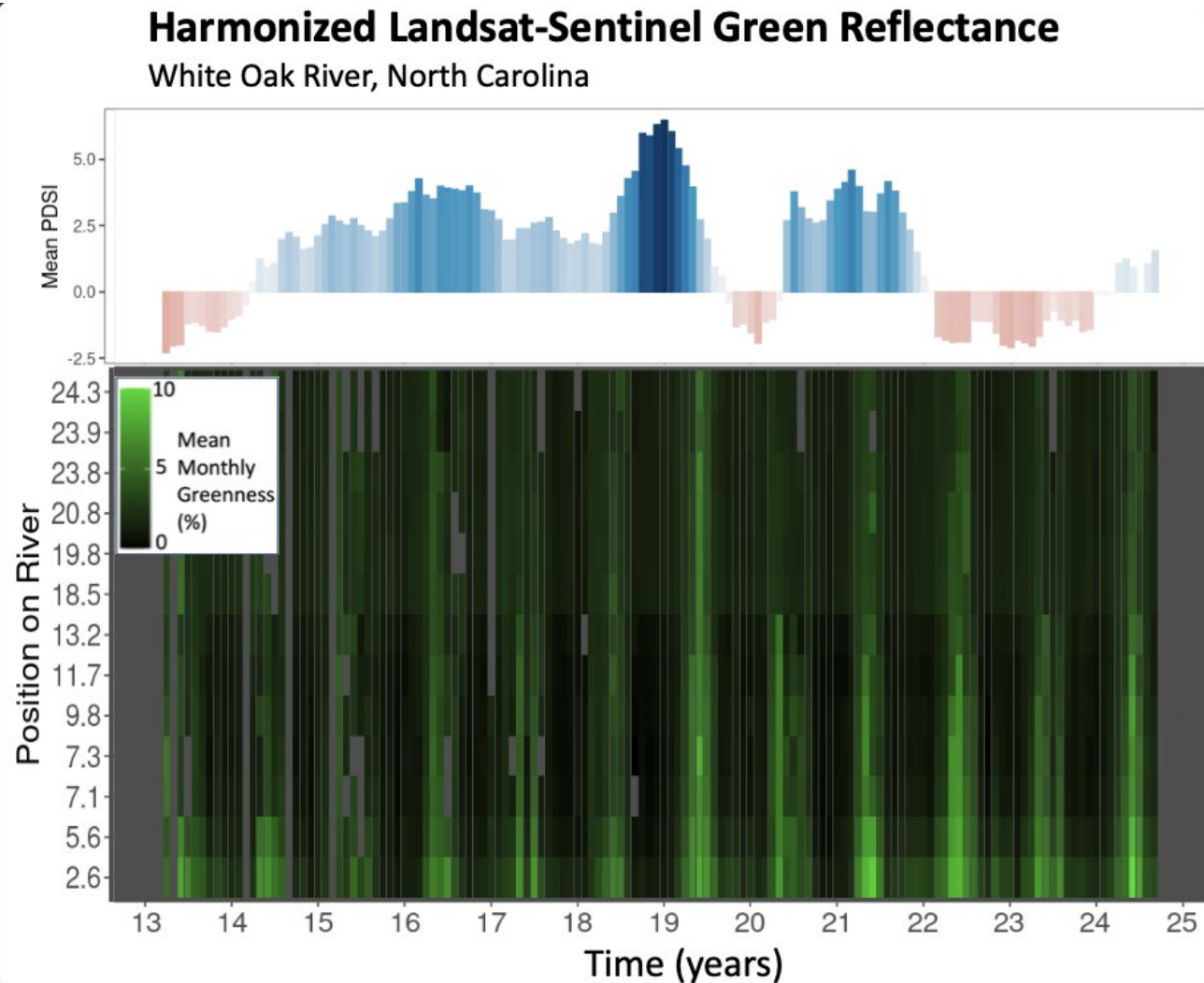
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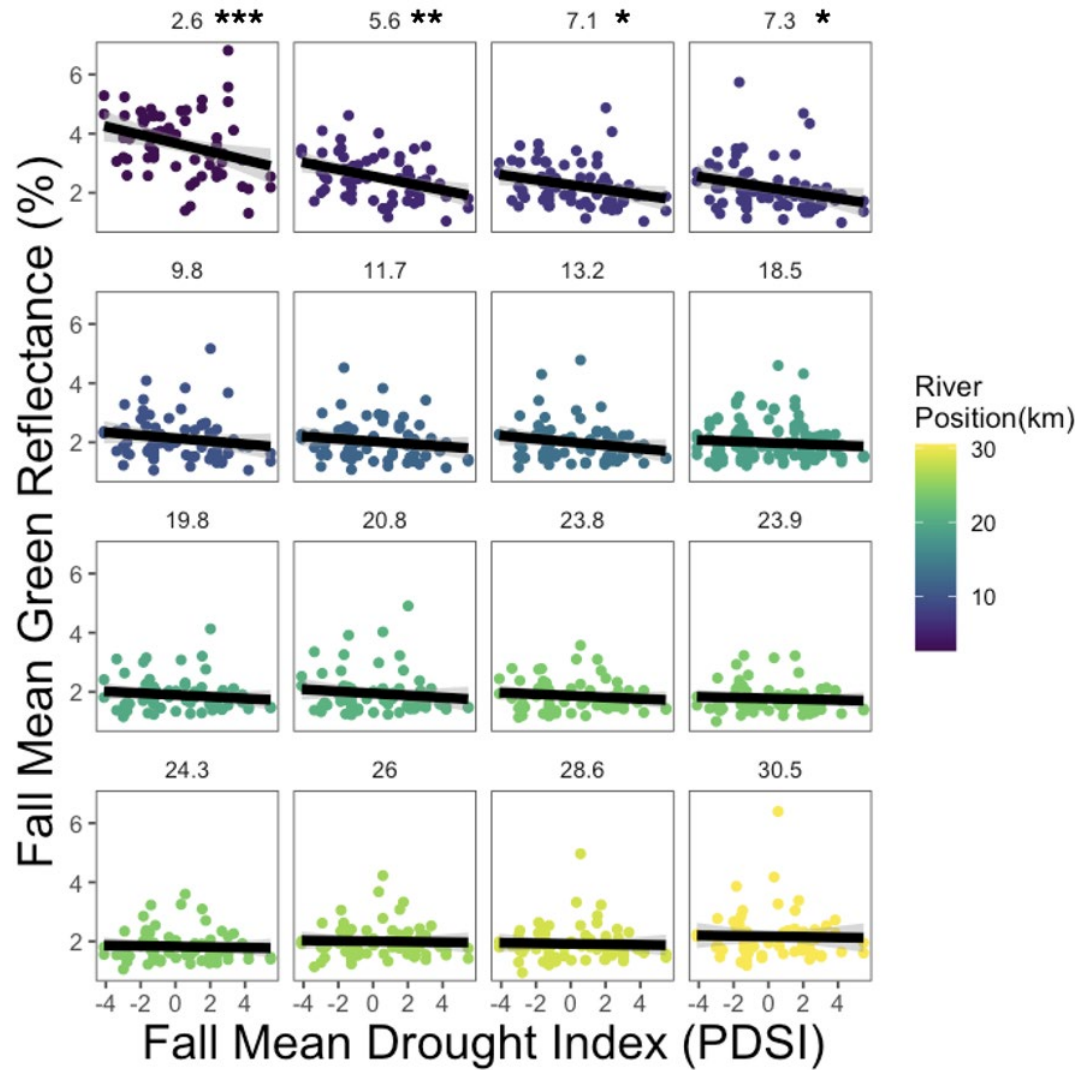
Can we answer our original questions?

2) What are the
consequences of
saltwater intrusion for
blackwater river
ecosystems?

We see a clear seasonal and spatial pattern in river greenness

Some evidence of saltwater intrusion impacting greenness







Thank you!

Southeast Rivers

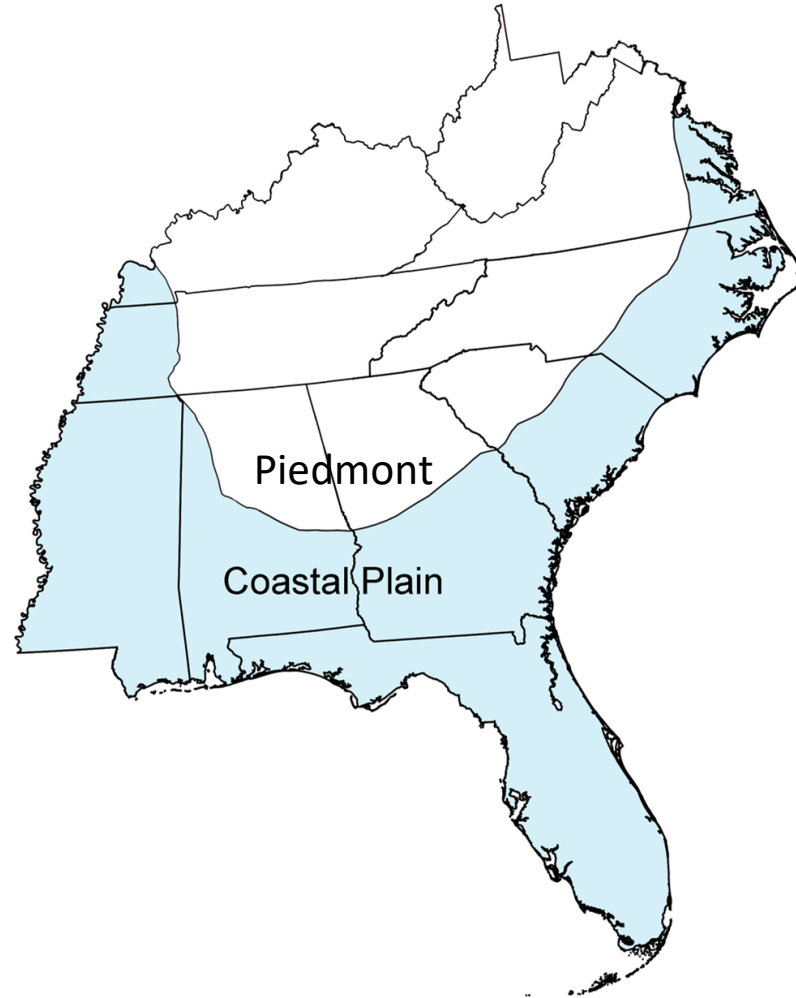
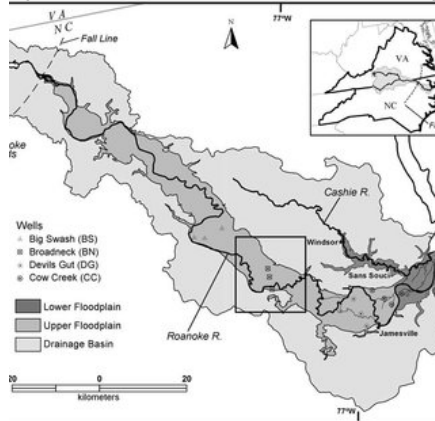
Two major underlying geologies:

- Coastal Plain: dominated by wetlands
- Piedmont: lots of clay

Roanoke River
(Piedmont dominated)



Cashie River
(Coastal plain dominated)



Roanoke vs Cashie rivers

Sentinel Image: January 21, 2023



What am I interested in:

- How does saltwater intrusion impact river color?
 - Does this differ between rivers draining different underlying geologies
- What are the long term trends in river color?
 - If happening, what is driving this change
 - Does this differ between rivers draining different underlying geologies

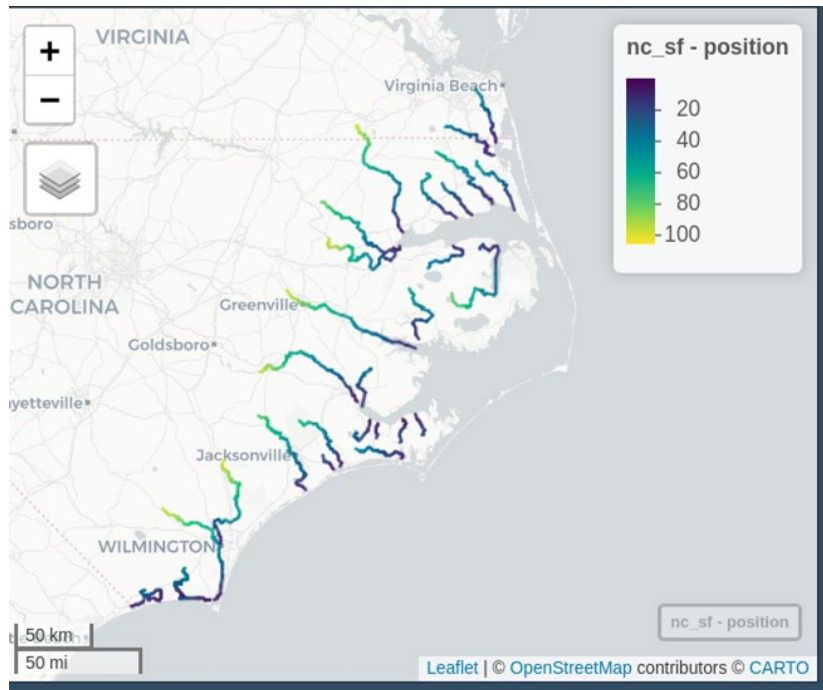
What I need help with:

- How should I approach these question in a manuscript
 - What should the scope be (geographic, both questions, etc.)?
 - Where to publish?
- What analyses/methods should I use
- How to communicate complicated geospatial and temporal data

Types of data

What I have:

- Landsat Surface Reflectance
 - 14 day return period
 - 1985 to present
- Harmonized Landsat-Sentinel Surface Reflectance
 - 3-1 data return period
 - 2017-present
- Palmer Drought Severity Index
 - Weekly gridded data product on drought severity
 - Interpolated to daily data aggregate by watershed
- Limited water quality data
 - Whatever was available on the National Water Quality Portal



How does saltwater
intrusion impact river
color?

White Oak River

nc_sf - position



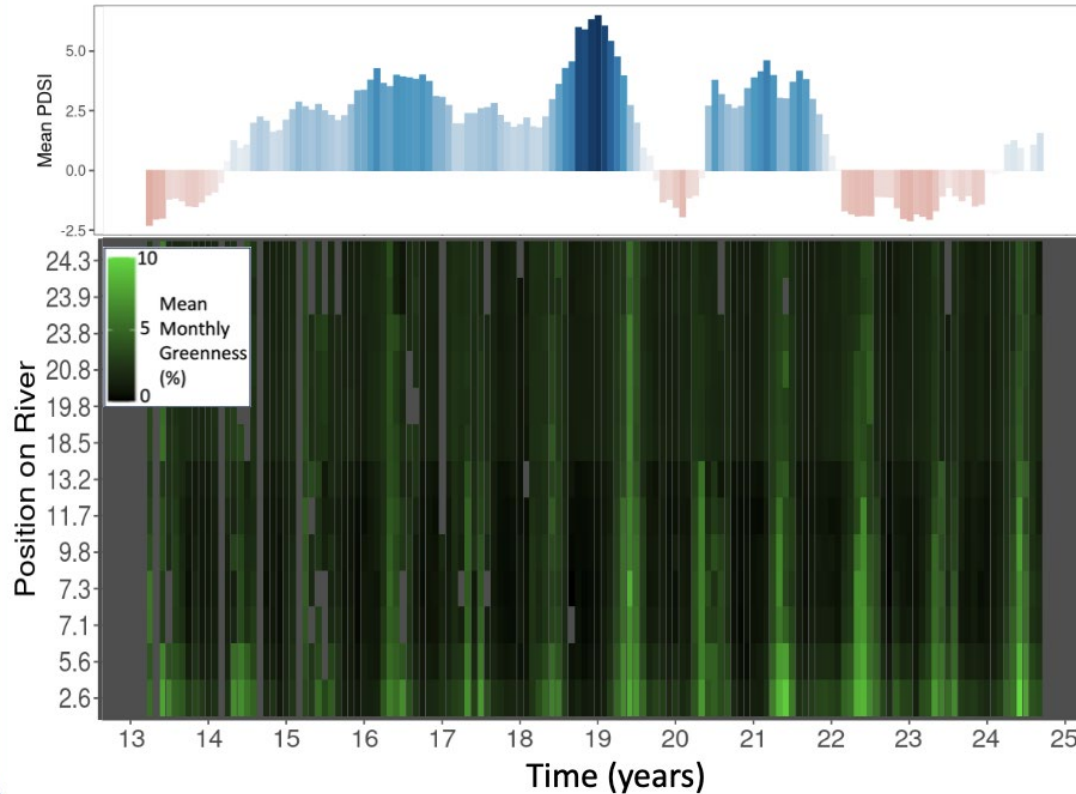
10 km

5 mi

Drought as saltwater intrusion proxy

Harmonized Landsat-Sentinel Green Reflectance

White Oak River, North Carolina

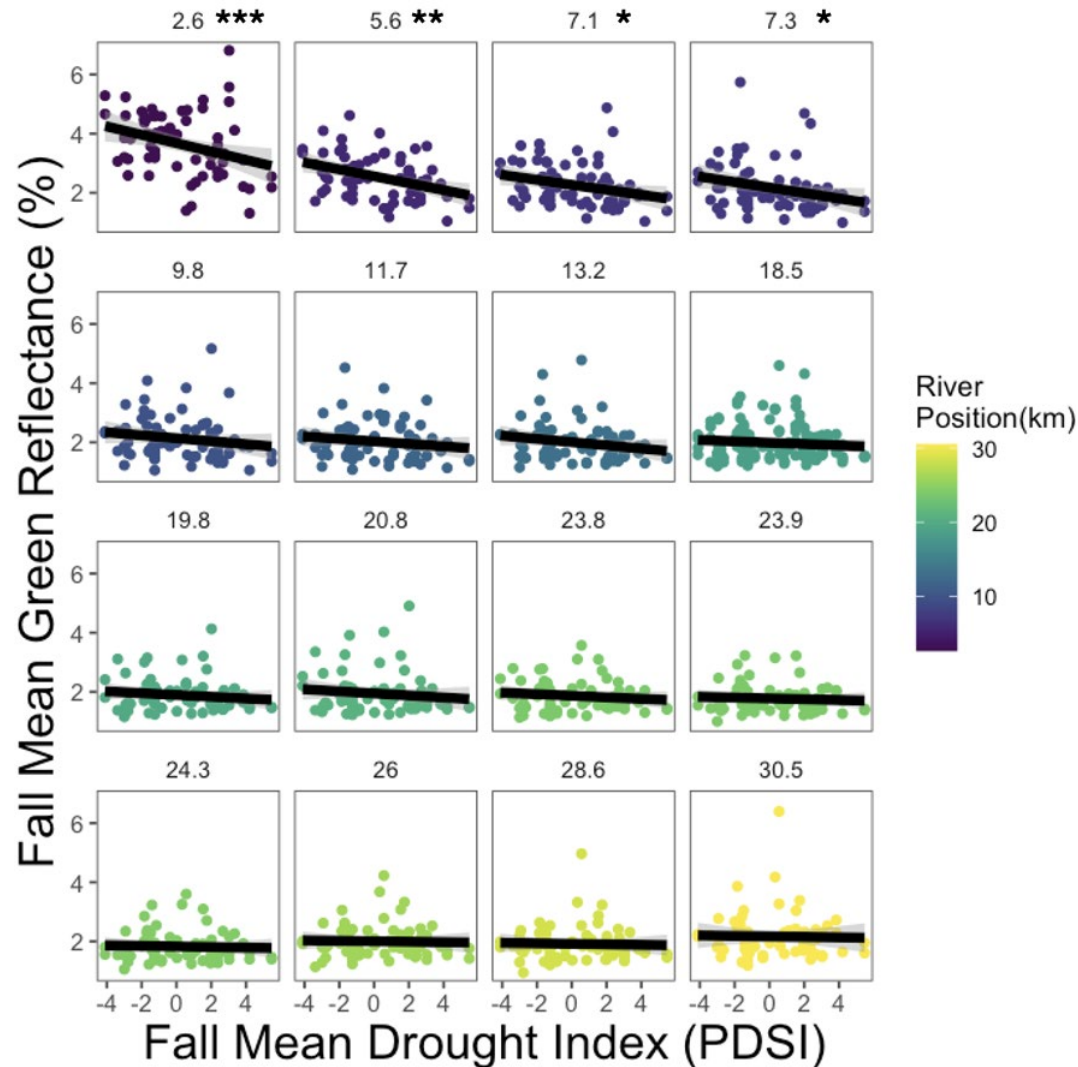


Drought as saltwater intrusion proxy

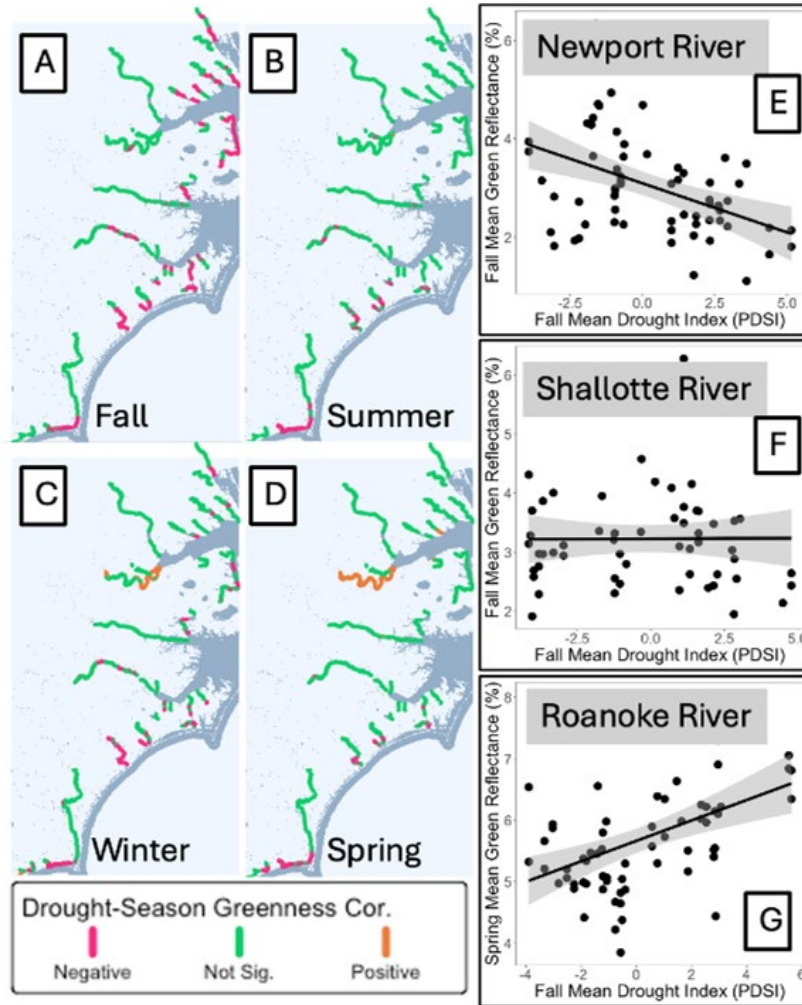
Landsat 5, 7, 8, 9 Surface Reflectance

Possible explanations:

- Clearing of water, increasing bottom reflectance (Likely)
- Increased light availability leading to increased productivity and chl-a contractions

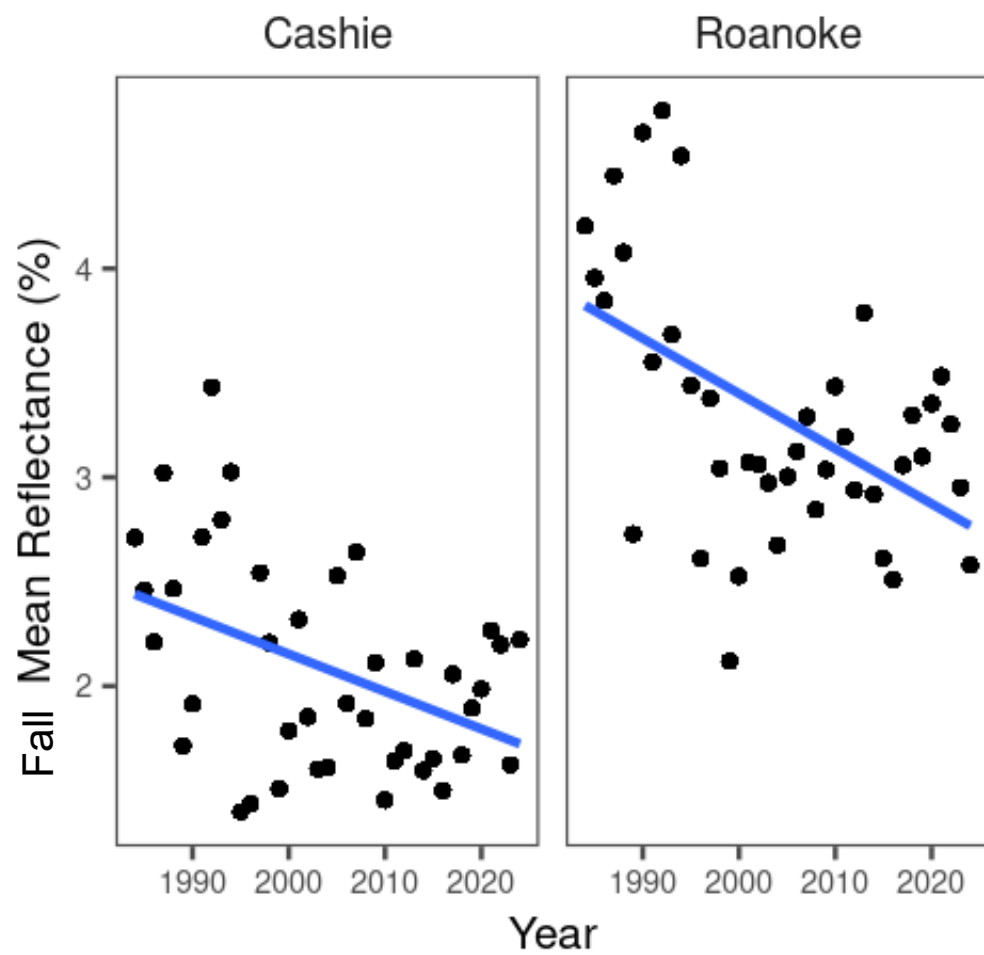


Region Relationships

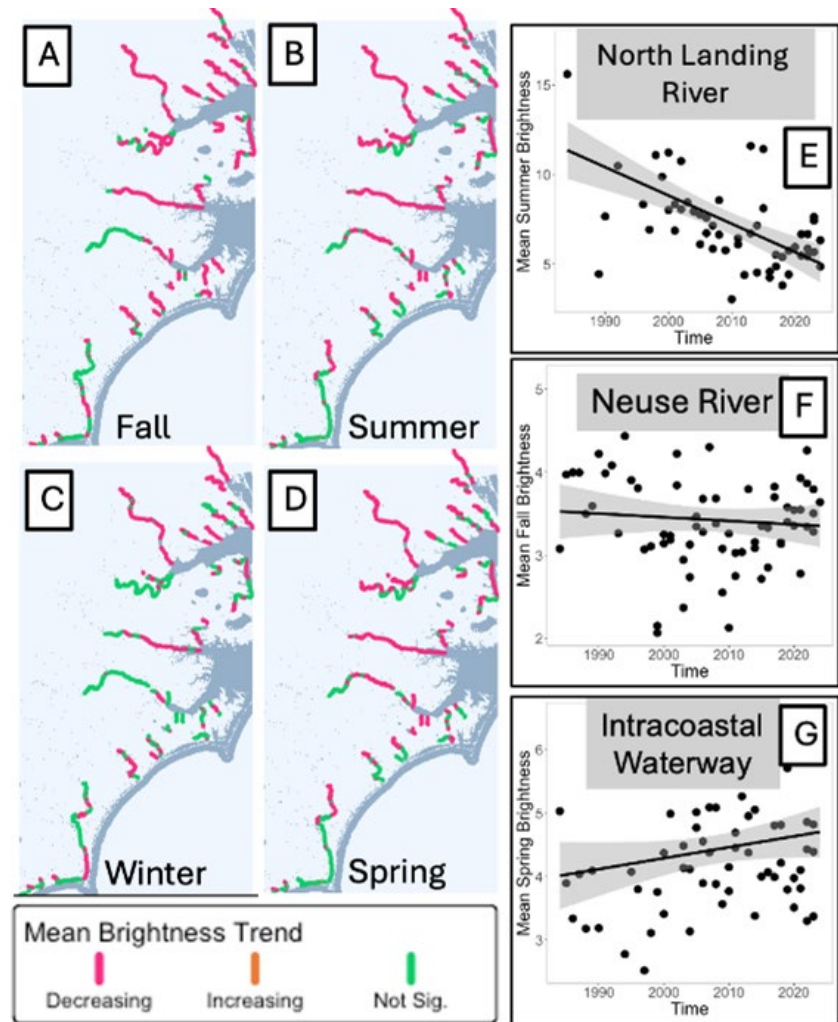


What are the long term
trends in river color?

Specifically brightness



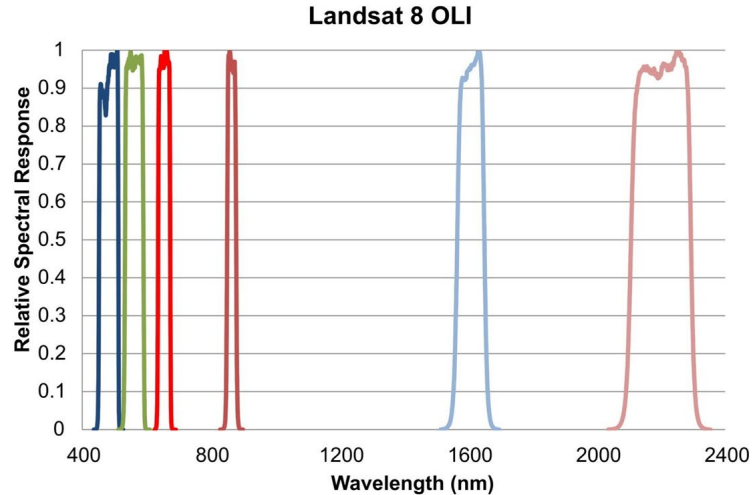
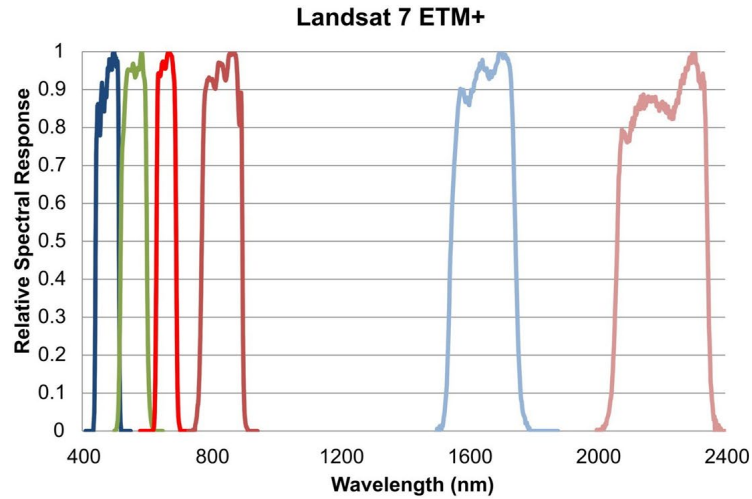
Region Relationships



Potential Issue:

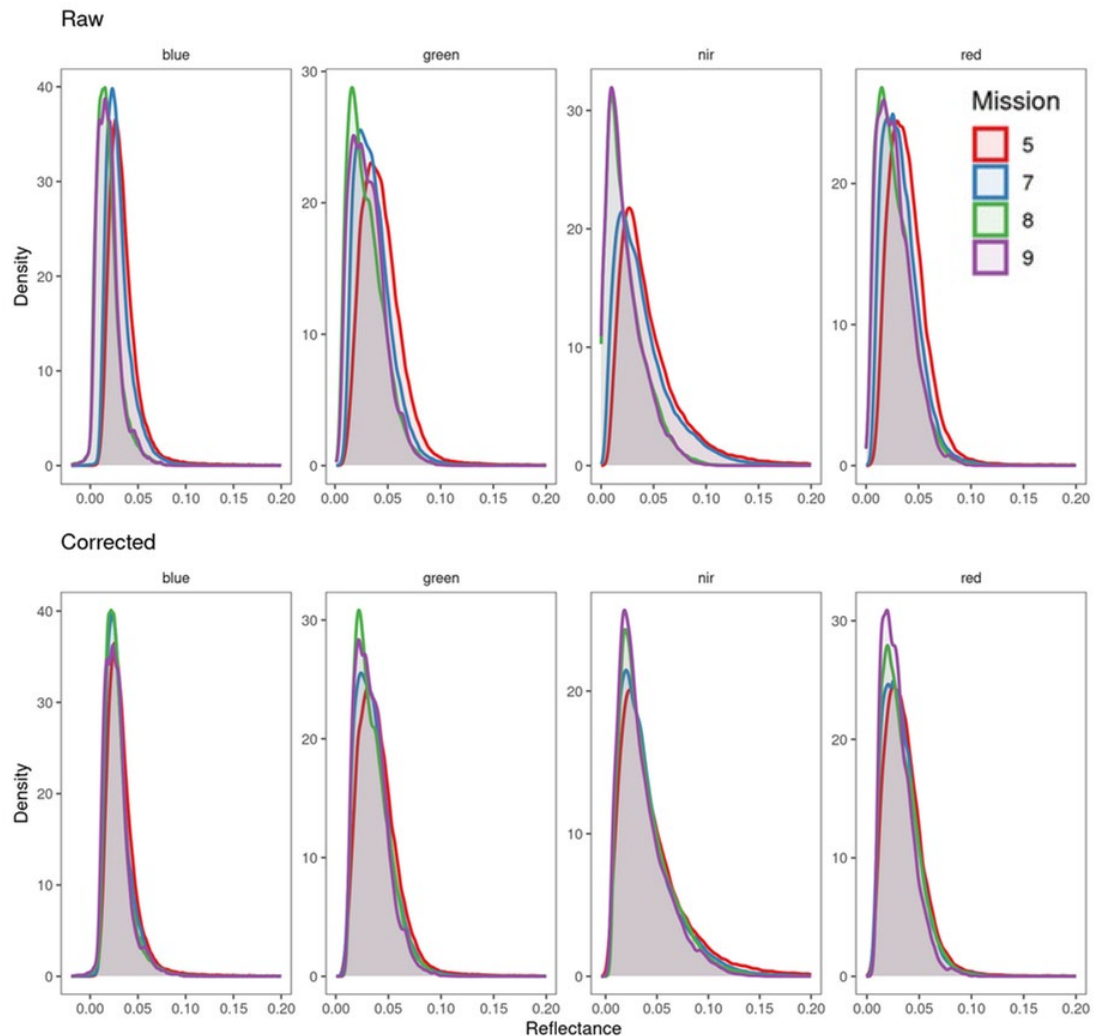
Differences between sensors

Landsat has 5 different satellites with similar but slightly different sensors



Corrections

How do you assess if a trend is real or due to an artifact like different sensors?



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Take away

- Questions:
 - Long term trend
 - Then saltwater
- Scope (geography)
 - Coastal plain rivers
 - Virginia to Georgia
- Scope (variables)
 - Coastal plain v piedmont
 - Protected vs open outlet
- Approach
 - Look at changes in brightness but also color
 - See if change is related to brightening (sediment) or darkening (cdom)